

Using WG t-test instead of WG ANOVA to do ANOVA SPSS Hw

With all the different versions of and vendors for SPSS, you might have ended up with an SPSS that doesn't have the "GLM: Repeated Measures" module that is needed to do a WG ANOVA. But, there is a work-around if the version of SPSS you have doesn't have that module!

The thing is that a 2BG ANOVA and a BG t-test are slightly different mathematical approaches to comparing the DV means from the conditions of a 2WG research design! So, you can run a WG t-test and then easily convert the t-test results to "F-test results"

Look at the WG t-test handout to see how to run the analysis.

Here's how to convert the t-test results into the answers for the WG ANOVA homework questions...

WG ANOVA results

Descriptive Statistics						
	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
Actor age: Clark Kent	9	30.58	4.88	1.63	24.47	37.84
Actor age: Lois Lane	9	30.17	6.41	2.14	23.32	40.10
Valid N (listwise)	9					

Tests of Within-Subjects Effects					
Measure: MEASURE_1					
Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Character	0.728	1	0.728	0.027	
Error(Character)	215.001	8	26.875		.873
Total	215.729	9			

WG t-test results

Descriptive Statistics						
	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
Actor age: Clark Kent	9	30.58	4.88	1.63	24.47	37.84
Actor age: Lois Lane	9	30.17	6.41	2.14	23.32	40.10
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Paired Samples Test						
	Mean	Std. Deviation	Std. Error Mean	t	df	Sig. (2-tailed)
clark_age - lois_age	0.402	7.331	2.444	0.165	8	.873

Notice the ns, means and standard deviations all match !!

To compute the various reporting values for the WG F questions on the assignment:

Conversion	Value
ANOVA $F = t^2 = 0.165^2$	$\rightarrow 0.027$
ANOVA df effect	$\rightarrow \text{always} = 1$
ANOVA df error = t-test df	$\rightarrow 8$

All of which match the values from the ANOVA output!

The key formulas are:

- $F = t^2$ (square the t-value)
- **df effect = 1** (always, for a 2-group comparison)
- **df error = t-test df** (same as the paired t-test degrees of freedom)
- **MSE** comes from the ANOVA output